Sec: SR.IIT_*COSC(MODEL-A&APEX) | GTM-2 | Date: 15-02-26
Time: 3 Hrs | JEE-MAIN | Max. Marks: 300

KEY SHEET

PHYSICS

1	**B**	2	**D**	3	**A**	4	**B**	5	**B**
6	**A**	7	**B**	8	**B**	9	**B**	10	**D**
11	**D**	12	**C**	13	**C**	14	**B**	15	**A**
16	**D**	17	**C**	18	**D**	19	**B**	20	**A**
21	**10**	22	**24**	23	**12**	24	**23**	25	**4**

CHEMISTRY

26	**A**	27	**D**	28	**B**	29	**D**	30	**B**
31	**B**	32	**A**	33	**B**	34	**D**	35	**B**
36	**C**	37	**C**	38	**D**	39	**C**	40	**A**
41	**B**	42	**B**	43	**D**	44	**C**	45	**A**
46	**5**	47	**48**	48	**7**	49	**5**	50	**50**

MATHEMATICS

51	**D**	52	**B**	53	**A**	54	**B**	55	**C**
56	**C**	57	**D**	58	**C**	59	**C**	60	**A**
61	**B**	62	**B**	63	**B**	64	**D**	65	**C**
66	**C**	67	**A**	68	**A**	69	**D**	70	**A**
71	**10**	72	**4**	73	**9**	74	**14**	75	**0**

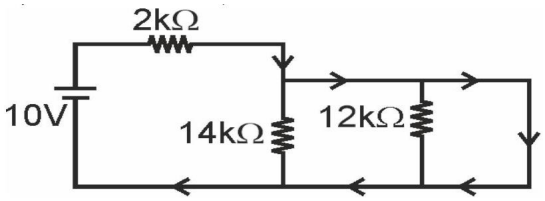
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SOLUTIONS
PHYSICS

1. For same deviation i & e are interchanged. $\delta = (i+e) - A$

$A = (i+e) - \delta \;\; A = (17+39) - 27 \;\; A = 56 - 27 = 29^\circ$

2. In Forward bias, ideal diode behaves as a short circuit



14kΩ & 12kΩ are shorted

so $I = \frac{10}{2} = 5$ mA

3. $\frac{dA}{dt} = \frac{L}{2m} =$ constant

Since, L = constant

mvr = constant

as r decreases from aphelion to perihelion, v increases.

4. $\vec{v} = x^2\hat{i} + 2y^2\hat{j}$

$v_x = x^2; v_y = 2y^2 \Rightarrow \frac{dx}{dt} = x^2; \frac{dy}{dt} = 2y^2 \therefore \frac{dx}{dy} = \frac{x^2}{2y^2} \Rightarrow \int_4^x \frac{dx}{x^2} = \int_1^y \frac{1}{2}\frac{dy}{y^2} \Rightarrow -\int_4^x \frac{1}{x}dx = -\frac{1}{2}\int_1^y \frac{1}{y}dy$

$\Rightarrow \frac{1}{x} - \frac{1}{4} = \frac{1}{2}\left[\frac{1}{y} - 1\right] \Rightarrow \frac{1}{x} - \frac{1}{2y} = -\frac{1}{4} \Rightarrow \frac{1}{2y} - \frac{1}{x} = \frac{1}{4}\; xy - 2x + 4y = 0$

5. $F = At + \frac{A}{x + Bm}$

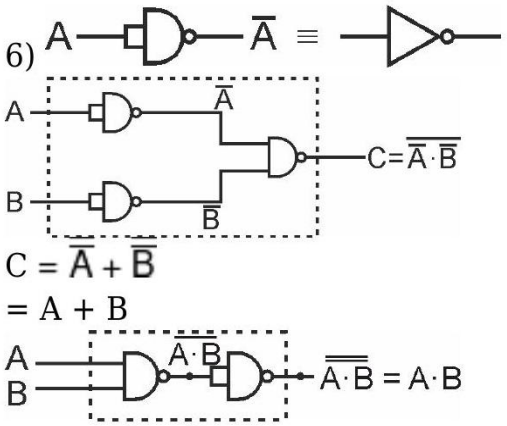
$[At] = [F]$

$\Rightarrow [AT] = [MLT^{-2}] \Rightarrow [A] = [MLT^{-3}] [x] = [Bm] \Rightarrow [L] = [BM] \Rightarrow [B] = [LM^{-1}]$

$\therefore \left[\frac{A}{B}\right] = \left[\frac{MLT^{-3}}{LM^{-1}}\right]$

$[M^2\ T^{-3}]$

6.



$C = \overline{A} + \overline{B}$

$= A + B$

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8. Taking out of page as +ve direction.

$L = mvR - I_{cm}\omega$

for ring $I_{cm} = mR^2$

$L^{=mvR} - mR^2\frac{2V}{R} = -mvR$

$L < 0$

as L is into the page, the ring will move towards right

For Solid sphere $I_{cm} = \frac{2}{5}mR^2$

$L_L = mvR - \frac{2}{5}mR^2\frac{2v}{R}$

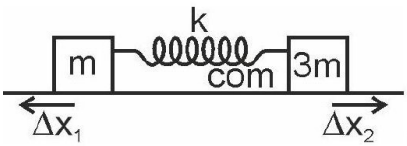
$L_L = mvR - \frac{4mvR}{5}$

$L_L = \frac{mvR}{5} > 0$

out of page

as L is out of the page, the sphere will move towards left.

10.



as $\vec{F}_{ext} = 0$

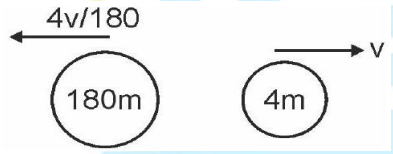
$\Delta r_{com} = 0$

$3\,m(\Delta x_2) + m(-\Delta x_1) = 0 \Rightarrow 3\Delta x_2 = \Delta x_1$

$\Delta x_1 + \Delta x_2 = \Delta x_1 + \frac{\Delta x_1}{3} = \ell$

$\Delta x_1 = \frac{3\ell}{4}$

11.



$\frac{1}{2}4mv^2 + \frac{1}{2}(180\ m)\left(\frac{4v}{180}\right)^2 = 5.5MeV$

$\frac{1}{2}4mv^2\left[1 + 45\left(\frac{4}{180}\right)^2\right] = 5.5MeV$

$KE_\alpha = \frac{5.5}{1 + 45\left(\frac{4}{180}\right)^2}MeV = 5.38MeV$

12. $V_1 = \frac{45-25}{\rho_1}$ and $V_2 = \frac{45-27}{\rho_2}$

$\frac{V_1}{V_2} = \frac{20}{18}\frac{\rho_2}{\rho_1} \Rightarrow \frac{V_1}{V_1(1+\gamma_m \times 10)} = \frac{20}{18}\frac{\rho_2}{\rho_1}$

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$$\frac{V_1}{V_1(1+\gamma_m\times 10)}=\frac{20}{18}\frac{\left(1.25\times 10^3\right)}{\left(1.5\times 10^3\right)}$$

$$\gamma_m=\frac{2}{250}$$

$$\gamma_m=3\alpha_m,\alpha_m=\frac{\gamma_m}{3}$$

On calculation

$$\alpha_m=2.6\times 10^{-3}/C$$

13. For first minimum a $\sin\theta=l$

So $\frac{\lambda}{a}=\sin 30^\circ=\frac{1}{2}$

Now for first secondary maximum a $\sin\theta=\frac{3\lambda}{2}$

Therefore $\sin\theta=\frac{3\lambda}{2a}=\frac{3}{2}\left(\frac{1}{2}\right)=\frac{3}{4}$

$$\Rightarrow\theta=\sin^{-1}\left(\frac{3}{4}\right)$$

14. $i_{rms}^2=\frac{1}{T}\int_0^T i^2dt, i=5t=\frac{1}{10}\int_0^{10}(5t)^2dt$

15. $T=2\pi\sqrt{\frac{\ell}{2g}}=2\pi\sqrt{\frac{1}{4g}}=1\text{sec}$

19. Direction of propagation,

$\hat{c}=\hat{E}\times\hat{B}=\hat{i}\times\hat{k}=-\hat{j}$

20. We know for paramagnetic material

$$\chi=\frac{C}{T}$$

$$\frac{1}{H}=\frac{C}{T}$$

$$I=\frac{CH}{T},H=\frac{B_0}{\mu_0}$$

$B_0=$ external magnetic field

$$l\propto\frac{B}{T}$$

$$\frac{I_1}{I_2}=\frac{B_1\times T_2}{T_1\times B_2}=\frac{0.8\times 20}{5\times 0.4}$$

$$\frac{I_1}{I_2}=8\Rightarrow I_2=\frac{I_1}{8}=\frac{0.8}{8}\times 0.1\ A/m$$

21. For plastic plate charge will be distributed on a single surface

So, $\sigma=\frac{Q}{A}\varepsilon_\varepsilon E_1=\frac{\sigma}{2\varepsilon_0}=\frac{Q}{2A\varepsilon_0}=10$

in copper plate, charge will be distributed on both the surface

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so $\sigma A \times 2 = Q \Rightarrow \sigma = \dfrac{Q}{2\ A}$

Electric field due to copper plate $= \dfrac{\sigma}{2\varepsilon_0} \times 2 = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{2\ A\varepsilon_0} = 10\ V/m$

22. By first polaroid P_1 intensity will be halved then P_2 and P_3 will make intensity $\cos^2(60^\circ)$ and $\cos^2(60^\circ)$ times respectively.

Intensity out $= \dfrac{256}{2} \times \dfrac{1}{4} \times \left(\dfrac{\sqrt{3}}{2}\right)^2 = \dfrac{256 \times 3}{2 \times 4 \times 4} = 24$

23.

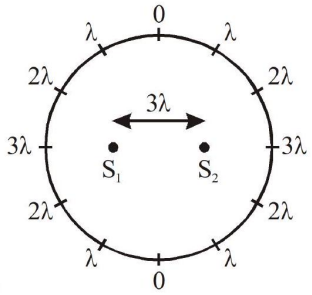
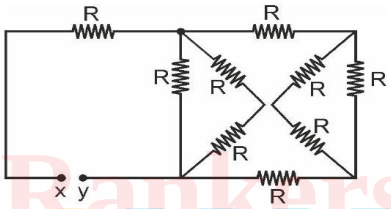


Diagram shows Δx at different point on circle From diagram it is clear that number of maxima $= 12$

24.



$$R_{eq.} = R + \frac{\frac{2R}{3} \times \frac{8R}{3}}{\frac{2R}{3} + \frac{8R}{3}} = R + \frac{8R}{15} = \frac{15R + 8R}{15}$$

$$= \frac{23R}{15} = 23 \times \frac{15}{15} = 23$$

25. $E_{2n} - E_1 = 204 \Rightarrow \dfrac{E_1}{4n^2} - E_1 = 204$

$E_1\left[\dfrac{1}{4n^2} - 1\right] = 204 \#(i)$ and $E_{2n} - E_n = 40.8$

$E_1 \times \dfrac{-3}{4n^2} = 40.8 \#(ii)$

From (i) & (ii)

$n = 2$

& $E_1 = -217.6eV$

$E_1 = -13.6Z^2$

$\dfrac{217.6}{13.6} = Z^2$

$Z = 4$

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CHEMISTRY

26. $\Delta T_b = i.K_b.m$

$10.4 = 4 \times 0.52 \times m$

$m = \dfrac{10.4}{4 \times 0.52} = 5$

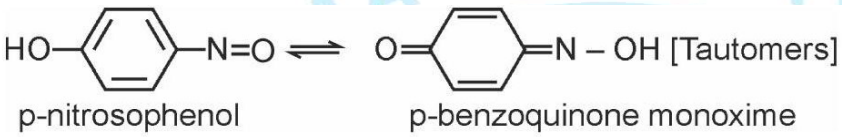
$m = \dfrac{n_{solute}}{w_{solvent(Kg)}}$

$\Rightarrow n_{solute} = 5 \times 0.25 = 1.25$

$m = 40$

27. NCERT statement.

28. Nitrous acid gives $\overset{\oplus}{N}O$ ion $\left(E^{\oplus}\right)$ which attacks phenol at less hindered p-position leading to formation of p-nitrosophenol.



p-nitrosophenol p-benzoquinone monoxime

29. Due to common ion effect, dissociation of both salt is suppressed. Hence each salt dissolves less than it would in pure water.
So simultaneous solubility < individual solubility.

30. $FeSO_4 \cdot (NH_4)_2 SO_4 \cdot 6H_2O (Mw 392gm)$

1 mol of this compound contain $= 64gm$ of ' S ' present along with $2\ mol NH_4^+$ in solution

$n_s = \dfrac{6.4 \times 10^{-3}}{32} = 2 \times 10^{-4}\ mol$

$n_{NH_4^+} = 2 \times 10^{-4}\ mol$

Since the amount of solute low dsolution $= 1gm / ml$

Molarity of $NH_4^+ = \dfrac{2 \times 10^{-4}}{0.1} = 2 \times 10^{-3}$

31. First check central atom 3d < 4d < 5d Co → belongs to 3d
Rh → belongs to 4d
Ir → belongs to 5d
then check ligand strength
$Cl^- < CN^-$
strong ligand

32. Theory Based.

33. $A^{2+} \rightarrow A^{4+} + 2e^-\ E^\circ = -0.12\ V$

$A^{2+} + 2e^- \rightarrow A\ E^\circ = -0.15\ V$

$2\ A^{2+} \rightarrow A^{+4} + A\ E^\circ\ cell$

$= -0.15\ V - 0.12\ V = -0.27\ V$

$E^\circ_{cell} = -ve$

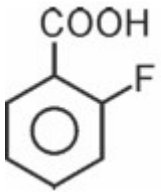
Non-spontaneous.

$E^\circ\ cell = 0.15\ V$

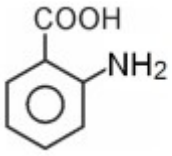
$A \rightarrow A^{4+} + 4e^-\ E^\circ\ cell = 0.03\ V$

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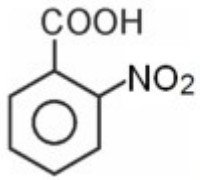
35.



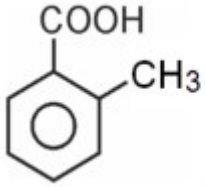
pKa = 3.3



pKa = 4.95



pKa = 2.17



pKa = 3.19

36. Last line of lyman series $\infty \to 1$

$\Delta E_n = E^o$

Δ En for $= \frac{99}{100}E^0 = E^0 \times \left(\frac{1}{1^2} - \frac{1}{n^2}\right)$

$1 - \frac{1}{100} = \frac{1}{1} - \frac{1}{n^2}$

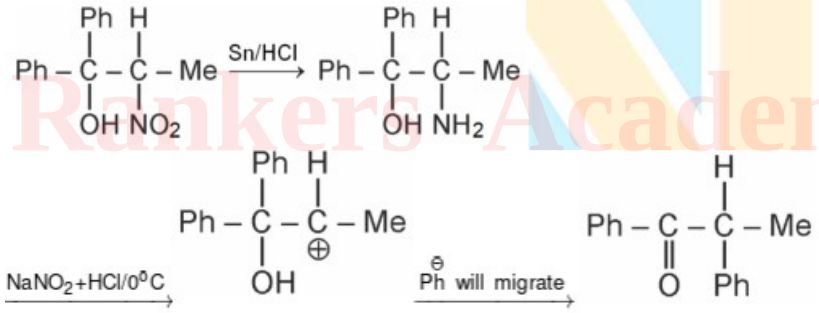
$n = 10$

no. of Paschen line $= 8$ to 3

$\Rightarrow 8 - 3 = 5$

37. Fe^{+2} Turnbull's Blue ppt.

38.



39. (i) $\left(M_{avg}\right)_{initial} = \frac{28+32}{2} = 30$

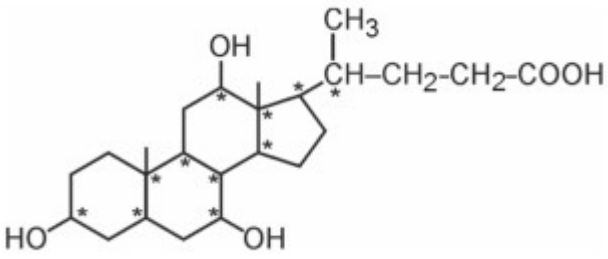
$(VD)_{initial} = \frac{30}{2} = 15$

(ii) $2CO + O_2 \to 2CO_2$

i	1	1	-
r	1	0.5	1
f	0	0.5	1

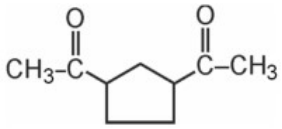
$\left(M_{average}\right)_{final} = \frac{0.5\times 32 + 1\times 44}{1.5} = \frac{16+44}{1.5} = 40$

40.

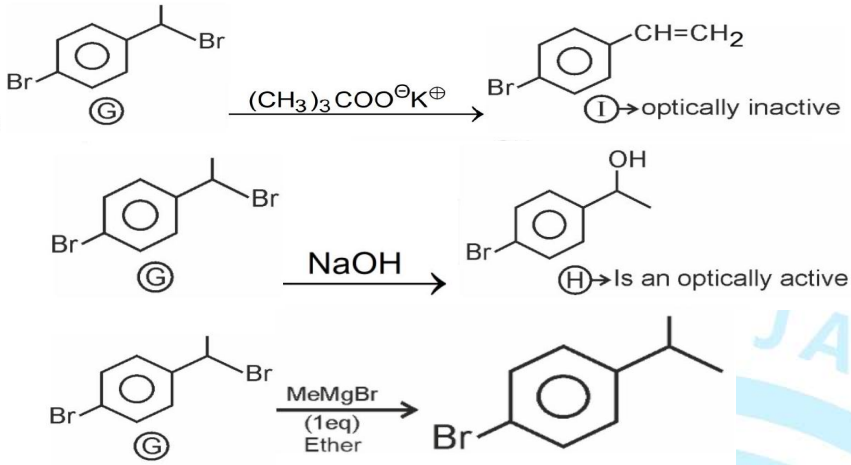


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41.



42)



43. No. of unpaired $e^- = 3\mu = \sqrt{n(n+2)}$

46. Compound :- $1,2,3,5,9$ are aromatic.

47. $2\,A(g) + B(D) \rightarrow 2C(g) + D(g)$

$\Delta_{ng} = 1$

$\Delta_r H = \Delta_r E + \Delta_{ng} RT = -20 + \dfrac{(1)\times 8\times 300}{1000}$

$\Delta_r H = -20 + 2.4 = -17.6\ kJ/mol$

$\Delta_r H = E_{af} - E_{ab}$

$E_{ab} = 30 - (-17.6) = 47.6\ kJ$

48. $Pb(NO_3)_2 + 2KI \rightarrow PbI_2(s) + 2KNO_3(aq)$

50 milli 200 milli

100 milli

$[I^-] = \dfrac{100}{500} = 0.2$

$(Pb^{+2}) = \dfrac{K_{SP}}{(I^-)^2} = \dfrac{8\times 10^{-9}}{(0.2)^2} = 2\times 10^{-7}\,M$

49. NCERT Based

50. gm equivalent of $KClO_3 = \dfrac{I.f.}{96500}$

$3\times 10^{-4} = 10\times 10^{-3} \times \dfrac{x}{100} \times \dfrac{9.65\times 60\times 60}{96500}$

$x = 50\%$

MATHEMATICS

51. Apply LMVT in $f'(x), x \in [-4,0]$

$\dfrac{f'(0) - f'(-4)}{4} = f''(C_1) \geq -3$

$f'(0) \geq -2 \#(i)$

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Apply LMVT in $f'(x), x \in [0,4]$

Also $\frac{f'(4)-f'(0)}{4} = f''(C_2) \geq -3$

$f'(0) \leq -2 \# (ii)$

from (i) & (ii) we get $f'(0) = -2$

Now again apply LMVT in $f'(x), x \in [0,x]$

$\frac{f'(x)-f'(0)}{x-0} \geq -3 \Rightarrow f'(x) \geq -2-3x$

Also by LMVT in $f'(x)$, when $x \in [x,4]$

$\frac{f'(4)-f'(x)}{4-x} \geq -3 \Rightarrow f'(x) \leq -14+3(4-x), \forall x \in (0,4)$

$\Rightarrow f'(x) < 0 \Rightarrow f(x)$ is decreasing for $\forall x \in (0,4)$

$\Rightarrow f(x) < 0; \forall x \in (0,4); f(0) = 0\ g'(x) = f(x) < 0; \forall x \in (0,4)$

$g(x)$ decreases in $(0,4)$

Also $g''(x) = f'(x) < 0$

$g(x)$ is concave down is $(0,4)$

52. Let $3^K = \alpha, 2^K = \beta$

$= \frac{6^K}{(3^K-2^K)(3^{K+1}-2^{K+1})} = \frac{\alpha \cdot \beta}{(\alpha-\beta)(3\alpha-2\beta)}$

$= \frac{\beta(3\alpha-2\beta)-2\beta(\alpha-\beta)}{(\alpha-\beta)(3\alpha-2\beta)} = \frac{\beta}{\alpha-\beta} - \frac{2\beta}{3\alpha-2\beta}$

Now, $\sum_{k=1}^{\infty} \frac{6^K}{(3^K-2^K)(3^{K+1}-2^{K+1})} = \sum_{k=1}^{\infty} \left(\frac{2^K}{3^K-2^K} - \frac{2^{K+1}}{3^{K+1}-2^{K+1}} \right)$

$= 2$

53. Equation of normal at point $P(10\cos\theta, 5\sin\theta)$ is

$10x\sec(\theta) - 5y\operatorname{cosec}(\theta) = 75$

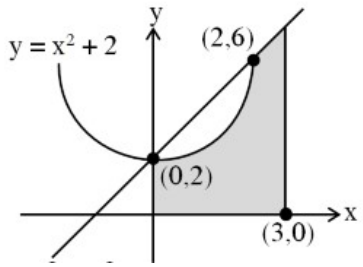
$2x\sec(\theta) - y\operatorname{cosec}(\theta) = 15$

Perpendicular distance from origin on normal is equal to radius of circle

$\Rightarrow r = \frac{15}{\sqrt{4\sec^2(\theta)+\operatorname{cosec}^2(\theta)}} \Rightarrow r = \frac{15}{\sqrt{45+4\tan^2(\theta)+\cot^2(\theta)}}$

$\Rightarrow r$ is maximum when $\tan^2(\theta) = \frac{1}{2}$

$\Rightarrow r_{max} = 5$



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54. $y = 2x + 2$

$A = \int_0^2 (x^2 + 2)\,dx + \int_2^3 (2x + 2)\,dx$

$A = \frac{41}{3}$

$12\ A = 41 \times 4 = 164$

55. Mean of $A = \mu_A$

Mean of $B = \mu_B = 2\mu_A$

Variance of $A = \sigma_A^2 = 16$

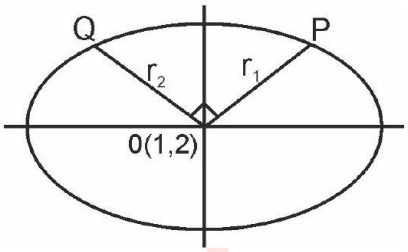
Variance of $B = \sigma_B^2 = 25$

Mean of combined data $= \mu_C = \frac{3\mu_A}{2}$

$\sigma_C^2 = \frac{1}{2}\left[\sigma_A^2 + (\mu_A - \mu_C)^2 + \sigma_B^2 + (\mu_B - \mu_C)^2\right]$

$= \frac{41}{2} + \frac{\mu_A^2}{4}$

56.



$E : \frac{(x-1)^2}{16} + \frac{(y-2)^2}{9} = 1$

$P(1 + r_1\cos\theta, 2 + r_1\sin\theta)$

$Q(1 + r_2\cos(90 + \theta), 2 + r_2\sin(90 + \theta))$

P & Q lies on the ellipse, then

$\frac{r_1^2\cos^2\theta}{16} + \frac{r_1^2\sin^2\theta}{9} = 1$

$\frac{1}{r_1^2} = \frac{\cos^2\theta}{16} + \frac{\sin^2\theta}{9}$

Similarly, we get

$\frac{1}{r_2^2} = \frac{\sin^2\theta}{16} + \frac{\sin^2\theta}{9} \Rightarrow \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{1}{16} + \frac{1}{9} = \frac{25}{16 \times 9}$

$\sqrt{\frac{1}{r_1^2} + \frac{1}{r_2^2}} = \frac{5}{12}$

57. Tangent at (x_1, y_1) on the parabola $y^2 = 4ax$

$yy_1 = 2a(x + x_1)$

any point on this tangent will be $\left(x', \frac{2a(x' + x_1)}{y_1}\right)$ with respect to circle

$x^2 + y^2 = a^2$ is $xx' + y.\frac{2a(x' + x_1)}{y_1} = a^2$

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$\Rightarrow xx'y_1 + 2ayx' + 2ayx_1 = a^2y_1$

$\Rightarrow \left(2ayx_1 - a^2y_1\right) + x'\left(xy_1 + 2ay\right) = 0$

Which is family of straight lines passing through point of intersection of $2ayx\ a_1 - a^2y_1 = 0$ and $xy_1 + 2ay = 0$

$\therefore y = \frac{ay_1}{2x_1}$ and $x = -\frac{2ay}{y_1}$

$\Rightarrow x = \frac{-a^2}{x_1}$

The fixed point is (x_2, y_2)

then $x_2 = \frac{-a^2}{x_1}$ & $y_2 = \frac{ay_1}{2x_1}$

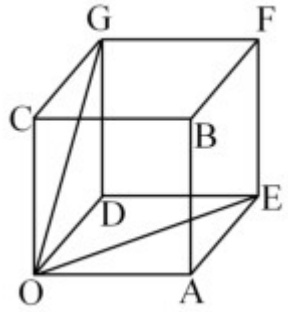
$$\therefore 4\left(\frac{x_1}{x_2}\right) + \left(\frac{y_1}{y_2}\right)^2 = 4\left(\frac{x_1}{-\frac{a^2}{x_1}}\right) + \left(\frac{y_1}{+\frac{ay_1}{2x_1}}\right)^2$$

$$= -\frac{4x_1^2}{a^2} + \frac{4x_1^2}{a^2} = 0$$

58.
$$= \frac{\sum\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)} = \frac{27-6\sum\left(\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}\right)+\frac{1}{\alpha\beta}+\frac{1}{\beta\gamma}+\frac{1}{\gamma\alpha}}{\left(3-\frac{1}{\alpha}\right)\left(3-\frac{1}{\beta}\right)\left(3-\frac{1}{\gamma}\right)}$$

$$= \frac{11}{7}$$

59.



(A) $P = \frac{{}^4C_3 \times 6 + 4 \times 2}{{}^8C_3} = \frac{4}{7}$

(3 vertices in single phase) + (selecting adjacent faces 1-2 or 2-3 or 3-4 or 4-5 and on side 5 or 6 is selected automatically)

$1 \to OAED, 2 \to AEFB, 3 \to BFGC,$

$4 \to GCOD, 5 \to DEFG, 6 \to OABC$

(B) $P = \frac{{}^4C_3 \times 6}{{}^8C_3} = \frac{3}{7}$

(C) $P = 1 - \frac{4}{7} = \frac{3}{7}$

(D) $P = \frac{{}^4C_3 \times 6 + 6 \times 4}{{}^8C_3} = \frac{48}{56} = \frac{6}{7}$

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60. $= \sum_{k=0}^{1012}\left(\frac{^{1012}C_k}{^{2024}C_k}\sum_{r=k}^{2024} {}^{r}C_k \cdot {}^{2024}C_r\right)$

$= \sum_{k=0}^{1012}\left({}^{1012}C_k \sum_{r=k}^{2024}\frac{{}^{r}C_k \cdot {}^{2024}C_r}{{}^{2024}C_k}\right)$

1012

$= \sum_{k=0}\left({}^{1012}C_k (2)^{2024-k}\right)$

$= 6^{1012} = (7-1)^{1012} = 49r + 22$

61. When $x \in \left(0, \frac{\pi}{4}\right), \cot(x) \in (1, \infty)$

$\Rightarrow \cot(x) < \cot^2(x) < \cot^3(x)$

and $x^2\langle x; -x^2\rangle - x$

$\Rightarrow I_4 > I_3 > I_2 > I_1$

62. $D_f : x \in \left[\sqrt{2}, 2\right] \cup \{0\}$

$\Rightarrow x = 0$ is not the solution

$\Rightarrow \forall x \in \left[\sqrt{2}, 2\right]; f(x)$

$f(x) = \sin^{-1}(x-1)^2 + \cos^{-1}\left(\frac{1}{x^2-1}\right)$

& $g(x) = \tan^{-1}(x)$ are strictly increasing functions,

where $^{f}\left(\sqrt{2}\right)\langle g\left(\sqrt{2}\right) \& f(2)\rangle g(2)$

so 1 solution.

63. Let $P\left(t, t^2\right)$ be the parametric co-ordinates of any point on the parabola $y = x^2$

$\Rightarrow \left.\frac{dy}{dx}\right|_{(t,t^2)} = 2t$

Hence normal at P is $\left(y - t^2\right) = \frac{-1}{2t}(x - t)$

If this line meets the curve again at $Q\left(t_1, t_1^{\,2}\right)$, then

$t_1^{\,2} - t^2 = \left(-\frac{1}{2t}\right)(t_1 - t)$

$\Rightarrow t_1 = -t - \frac{1}{2t}$

If □ is length of the normal chord then $z = (\blacklozenge)^2 = PQ^2$

$\Rightarrow Z = (t - t_1)^2\left[1 + (t + t_1)^2\right]$

$\Rightarrow z = \left(t + t + \frac{1}{2t}\right)^2\left(1 + \left(-\frac{1}{2t}\right)^2\right)$

$\Rightarrow z = \frac{\left(4t^2+1\right)^3}{16t^4}$

Let put $4t^2 = p$; where $p > 0$

$z = \frac{(p+1)^3}{p^2} = (p+3) + \frac{3}{p} + \frac{1}{p^2}$

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$\Rightarrow \frac{dz}{dp}=\left(1-\frac{3}{p^2}\right)-\frac{2}{p^3}=0$

$\Rightarrow (p+1)^2(p-2)=0$

$\Rightarrow p=2$ as $p\neq -1$

and $\frac{d^2z}{dp^2}=\frac{6}{p^3}+\frac{6}{p^4}>0$

Hence z is minimum when $p=2=4t^2$

64. Clearly domain is $[-1,1]$

$\sin^{-1}x+\tan^{-1}x$ is increasing in $[-1,1]$

and $\cos^{-1}x+\cot^{-1}x$ is decreasing in $[-1,1]$

So, maximum value of the function occurs at $x=\pm 1$

$$\left|\frac{\sin^{-1}1+\tan^{-1}1}{\cos^{-1}1+\cot^{-1}1}\right|=\left|\frac{\frac{\pi}{2}+\frac{\pi}{4}}{0+\frac{\pi}{4}}\right|=2+1=3$$

$$\frac{\sin^{-1}(-1)+\tan^{-1}(-1)}{\cos^{-1}(-1)+\cot^{-1}(-1)}\Bigg|=\left|\frac{-\frac{\pi}{2}-\frac{\pi}{4}}{\pi+\frac{3\pi}{4}}\right|=\frac{\frac{3\pi}{4}}{\frac{7\pi}{4}}=\frac{3}{7}$$

Clearly maximum value is 3 .

65. $\left(\left(\hat{a}+\hat{b}\right)+2\left(\hat{a}\times\hat{b}\right)\right)\cdot\left(\left(\hat{a}+\hat{b}\right)+2\left(\hat{a}\times\hat{b}\right)\right)=4$

$\left|\hat{a}+\hat{b}\right|^2+4\left|\left(\hat{a}\times\hat{b}\right)\right|^2+0=4$

Let the angle be θ between $\hat{a}$ and $\hat{b}$

$2+2\cos\theta+4\sin^2\theta=4$

$2+2\cos\theta-4\cos^2\theta=0$

Let $\cos\theta=t$ then

$2t^2-t-1=0$

$2t^2-2t+t-1=0$

$2t(t-1)+(t-1)=0$

$(2t+1)(t-1)=0$

$t=-\frac{1}{2}$ or $t=1$

$\cos\theta=-\frac{1}{2}$ not possible as $\theta\in(0,\pi)$

$\theta=\frac{2\pi}{3}$

Now,

$S_1\ 2\left|\vec{a}\times\vec{b}\right|=2\sin\left(\frac{2\pi}{3}\right)$

$\left|\hat{a}-\hat{b}\right|=\sqrt{1+1-2\cos\left(\frac{2\pi}{3}\right)}$

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$=\sqrt{2-2\times\left(-\frac{1}{2}\right)}=\sqrt{3}$

S_1 is correct

S_2 projection of â on $\left(\hat{a}+\hat{b}\right)$

$$\frac{\hat{a}.\left(\hat{a}+\hat{b}\right)}{\left|\hat{a}+\hat{b}\right|}=\frac{1+\cos\left(\frac{2\pi}{3}\right)}{\sqrt{2+2\cos\frac{2\pi}{3}}}$$

$$=\frac{1-\frac{1}{2}}{\sqrt{1}}=\frac{1}{2}$$

C option is true.

66. Let, centre $\equiv(h,k)$

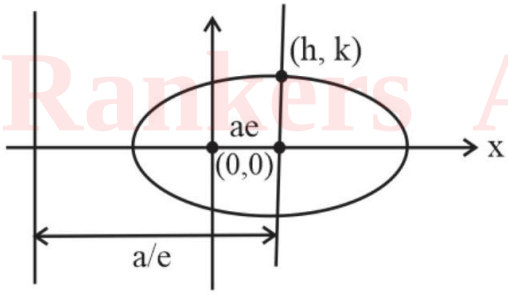
$$\left|\frac{5h+12k-10}{13}\right|=3 \text{ and } \left|\frac{5h-12k-40}{13}\right|=3$$

$\Rightarrow 5h+12k-10=\pm 39\#(i)$

$\Rightarrow 5h-12k-40=\pm 39\#(ii)$

Solving (i) & (ii)

$\Rightarrow h=5, k=2$



67. $x=-4$

$\Rightarrow k=b, h=ae$ and $a/e-ae=4$

$\Rightarrow a\left(1-e^2\right)=4e \Rightarrow a^2\left(1-e^2\right)=4ae$

$\Rightarrow b^2=4h$

$\Rightarrow k^2=4h \Rightarrow y^2=4x$

68. Let $\log_{30}\left(\log_{15}30\right)=t$

& $\left(\log_{30}\left(\log_{30}15\right)=\left(\log_{30}\left(\log_{15}30\right)^{-1}=-t\right.\right.$

$f(t)+f(-t)=12$

$f(-t)=5$

69. $\sum_{r=0}^{16}(-1)^{r16}C_r=2\sum_{r=0}^{6}(-1)^{r16}C_r-2.^{16}C_7+^{16}C_8=0$

$$\sum_{r=0}^{6}(-1)^{r\,16}C_r=^{16}C_7-\frac{^{16}C_8}{2}=^{15}C_9$$

$=5.7.11.13$

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70. Reflexive and symmetric are obvious.

for transitive consider $a=\frac{3}{4},\ b=\frac{1}{3},c=\frac{1}{8}$

71. $z=x+iy$

$x-2=\left|(x-7)+i(y+2)\right|$

$(x-2)^2=(x-7)^2+(y+2)^2$

$x^2+4-4x=x^2+49-14x+y^2+4+4y$

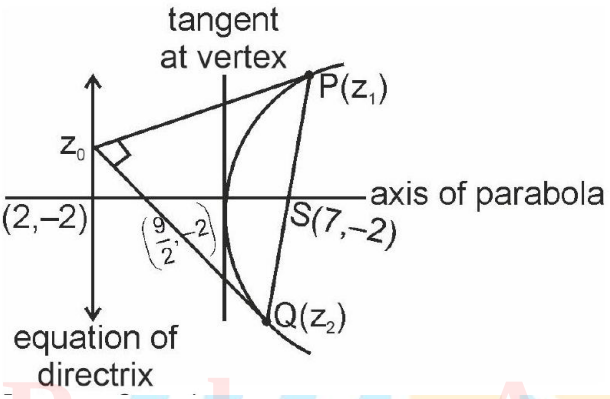
$10x-45=(y+2)^2$

$(y+2)^2=10(x-9/2)$

$y^2=4ax$

Focus $\equiv(7,-2)$

equation of directrix $\Rightarrow x=2$



Let $z_0=2+\alpha i$

PQ is focal chord

minimum value of $PQ=10$

72. $3xydy-2x^4dy-4x^3ydx+y^2dx=0$

Multiply with y

$\left(3xy^2dy+y^3dx\right)-\left(2x^4ydy+4x^3y^2dx\right)=0$

$d\left(xy^3\right)=d\left(x^4y^2\right)$

$xy^3=x^4y^2+c$

$\because f(1)=1\Rightarrow c=0$

$xy^3=x^4y^2$

$y=x^3$

$\frac{f(2)}{2}=4$

73. Symmetric matrix

$\Rightarrow A=\begin{pmatrix}a & b\\ c & a\end{pmatrix}$

Skew symmetric

$\Rightarrow A=\begin{pmatrix}0 & -\alpha\\ \alpha & 0\end{pmatrix}_{\text{type}}$

for Symmetric ($b=c$)

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$|A| = a^2 - b^2 = (a-b)(a+b) = 5\lambda$

Solve in the set $\{0,1,2,3,4\}$;

*a = b ; 5 solutions (one for each a).

$a+b \equiv 5\lambda$; 5 solutions (0,0), (1,4), (2,3), (3,2), (4,1)

So total solution $5+5-1=9$.

For Skew symmetric ($a=0, c=-b$)

$|A| = b^2 = 5\lambda$ so $b=0$

only zero matrix is a solution

so total number of solution are 9 .

74. $E_i \to i$ white and 10-i black

$A \to 3$ black balls are drawn

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)\cdot P\left(\frac{A}{E_1}\right)}{\sum P(E_i)\cdot\left(\frac{A}{E_1}\right)}$$

$$= \frac{\frac{1}{11}\times\frac{{}^9C_3}{{}^{10}C_3}}{\frac{1}{11}\left(\frac{{}^{10}C_3+{}^9C_3+\ldots+{}^3C_3}{{}^{10}C_3}\right)} = \frac{{}^9C_3/{}^{10}C_3}{{}^{11}C_4/{}^{10}C_3}$$

$$= \frac{14}{55}$$

75. Required plane will contain the points (1,2,3) and (−1,−3,1) and parallel to line having direction ratios $(2,3,4)$.

let equation of plane be $a(x-1)+b(y-2)+c(z-3)=0$

Now (−1,−3,1) also satisfy above equation

$a(-1-1)+b(-3-2)+c(1-3)=0$

$\Rightarrow -2a-5b-2c=0 \#(1)$

and $2a+3b+4c=0$

solving (1) and (2)

$$\frac{a}{-20+6} = \frac{b}{-4+8} = \frac{c}{-6+10} \#(2)$$

$$\Rightarrow \frac{a}{-7} = \frac{b}{2} = \frac{c}{2} = k(\text{ let })$$

so, $a=-7k, b=2k, c=2k$

therefore equation of plane is

$-7k(x-1)+2k(y-2)+2k(z-3)=0$

$\Rightarrow 7(x-1)-2(y-2)-2(z-3)=0$

$\Rightarrow 7x-2y-2z+3=0$

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